Open problems:
Unitary evolution as infinitesimal projective processes

Open problems raised by the technical brief in this directory, and by its Lean formalization (see `formalization-log.tex`). Items marked [BLOCKS VERIFICATION] stand in the way of checking a main claim of the brief as the brief states it: a *formalization gap* can be closed in Lean alone, while a *brief issue* needs the brief to be corrected or made precise first.

From the brief

1. **Thermodynamic analogy.** The brief compares the result with the way "reversible quasi-static evolution in thermodynamics can be understood as a limit of equilibration processes". No precise statement linking the two limits is given. (Brief, Sec. 1; Sec. 6: "a reversible quasi-static limit of projective processes".)

2. **Resolvent of an unbounded generator.** The brief states that "direct use of the resolvent of an unbounded generator still gives a completely positive trace-preserving map and the resolvent exponential formula", and that the spectral truncations are needed only for states outside the domain of H. Neither claim is proved, and whether the untruncated $(I - \tau\mathcal{L})^{-1}$ gives perfect prediction on the domain of H is left open. Not formalized. (Brief, Sec. 5, "Unbounded generators"; Log, Sec. 1.1, "Differences", last item.)

From the formalization

Beyond finite dimension the formalization has a part in any dimension, which covers unbounded H in DR-UNIT $\Rightarrow$ DR-PROJ; it leaves open the item "Resolvent of an unbounded generator" above and the items "DR-PROJ without `IsEvolution`", "Complete positivity", "The generator of the cutoff family" and "General cutoff levels" below (the second is open in finite dimension too).

1. [BLOCKS VERIFICATION: FORMALIZATION GAP; BRIEF ISSUE] **DR-PROJ without `IsEvolution`.** In infinite dimension Theorem 18 is proved only for families Φ_t that are already affine on density operators, with $\Phi_0 = I$, $\Phi_{t+s} = \Phi_t \circ \Phi_s$ and continuity at 0^+. The brief derives these from DR-PROJ (affinity of the limit, eqs. (28)–(29), common refinements, and "the continuity used in the proof is not an additional assumption"). In finite dimension they are now derived, with $\Phi_0 = I$ assumed, by the corrected argument of the next item (`IsProjectiveLimit.isEvolution`, `drUnit_iff_drProj_and_map_zero`); the argument uses a finite basis of density operators and does not carry over. (Log, "Differences", item on `IsEvolution`; Sec. 1.1, "Differences".)

2. [BLOCKS VERIFICATION: BRIEF ISSUE] **Durations with irrational ratio.** DR-PROJ defines Φ_t only as $\lim_n (\Psi_{t/n})^n$. For t/s irrational a common step τ gives blocks Ψ_τ^k that are not of

this form, so "common refinements" and continuity at zero do not prove $\Phi_{t+s} = \Phi_t \circ \Phi_s$; eq. (26) would have to hold uniformly along the trajectory. The remaining interval itself is harmless. A fix: in finite dimension a telescoping estimate gives $\Phi_t = e^{tL}$ directly, which is formalized (`IsProjectiveLimit.exists_eq_exp`); in infinite dimension Chernoff's product formula does, if the closure of L generates a contraction semigroup, which is open. (Brief, Sec. 4, after eq. (29); Log, "Problems found".)

3. [BLOCKS VERIFICATION: BRIEF ISSUE] **What perfect reconstruction means.** DR-PROJ only asks that the probabilities of the reconstructed sequences tend to one. The formalization instead assumes a second projective limit Φ'_t with $\Phi'_t(\Phi_t P) = P$ and $\Phi_t(\Phi'_t P) = P$ on pure states. It is not settled whether the brief's weaker wording already makes Φ_t a bijection of the pure states, or what the right formulation is. (Brief, Sec. 4: "Perfect reconstruction gives the converse correspondence"; Log, "Differences" and "Problems found".)

4. **Complete positivity.** Proposition 2 says that Ψ_τ is completely positive, as a mixture of unitary channels by eq. (7). Only positivity and trace preservation are formalized, in either dimension. (Log, "Differences"; Sec. 1.1, "Differences", last item.)

5. **The generator of the cutoff family.** For unbounded H the brief states that the family (23) is infinitesimal with generator $\mathcal{L}(\rho) = [H, \rho]/(i\hbar)$. The formalization only shows that $(\Psi_\tau(\rho) - \rho)/\tau$ converges on the core; it does not identify the limit. (Brief, Sec. 4, after eq. (23); Log, Sec. 1.1, "Differences", item on eq. (26).)

6. **General cutoff levels.** DR-UNIT $\Rightarrow$ DR-PROJ for unbounded H is formalized for the single choice $M_\tau = \tau^{-1/4}$. The brief's argument for every cutoff satisfying eq. (22) is not formalized. (Brief, eq. (22); Log, Sec. 1.1, "Differences".)

7. **The qubit picture.** $\Psi_\tau = (I + \mathcal{C}_\tau)/2$ with the Cayley transform $\mathcal{C}_\tau = (I + \tau\mathcal{L})(I - \tau\mathcal{L})^{-1}$, its orthogonality for the Hilbert–Schmidt inner product, and the midpoint picture on the Bloch ball are not formalized. (Brief, Sec. 5, "The qubit picture"; Log, "Problems found", last item.)

8. **The finite process is not unitary.** The claims that Ψ_τ is generally irreversible, maps pure ensembles to mixed ones, and that $I - \tau\mathcal{L}$ is not generally positive are not formalized. (Brief, Sec. 5, "The finite process is not unitary"; Log, "Problems found", last item.)